

MUNIM AHMAD

Full-Stack AI Engineer

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PROFILE

Full-Stack AI Engineer with expertise in building production-ready AI applications from model development to deployment. Specialized in RAG systems, LLM optimization, and scalable backend architecture. Proven track record of reducing model latency by 65% and delivering end-to-end ML solutions.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer — Full-Stack AI Development

May 2025 – October 2025

Endshift – Lahore, Pakistan

AI/ML Systems:

- Architected production RAG pipeline converting natural language to SQL using local LLMs, serving 500+ queries daily with 98% uptime
- Reduced Mistral-7B inference latency by 65% through advanced batching, quantization (4-bit), and optimization techniques
- Implemented vector embeddings and semantic search using ChromaDB; designed prompt engineering workflows with evaluation metrics

Backend & DevOps:

- Built scalable backend services using .NET (C#) with RESTful APIs; integrated Okta SSO for enterprise authentication
- Developed licensing/activation system with real-time validation; contributed to IPDR system with PostgreSQL optimization
- Deployed ML models on AWS EC2 with Docker; implemented Redis caching and monitoring for model performance tracking

TECHNICAL PROJECTS

Recallm – Python Semantic Cache Library for LLM Apps

<https://recallm.dev>

- Built and published open-source Python package (`pip install recallm`) that wraps OpenAI-compatible chat completion calls with in-process semantic caching
- Implemented local ONNX embedding + cosine similarity search with strict/balanced/loose threshold profiles, context-hash safeguards, TTL, and namespace-based invalidation
- Delivered production-focused reliability and visibility with sync/async support, Redis + thread-safe in-memory storage backends, fail-open behavior, Prometheus metrics, and Grafana dashboarding

Tech: Python, NumPy, FastEmbed (ONNX Runtime), SentenceTransformers, AsyncIO, Redis, ThreadSafeInMemoryStorage, Structlog, Prometheus, Grafana, Pytest, Ruff, MyPy

Snag – Open-Source Webhook Inspector, Tunnel & AI Agent Bridge

<https://snag-web-five.vercel.app/>

- Built a full-stack, open-source webhook platform with five integrated surfaces: a real-time Next.js web console, a Fastify capture server, a CLI tunnel, a Python MCP server, and a zero-dependency TypeScript SDK
- Implemented WebSocket-based live request feed with replay, JSON diff viewer, and forwarding rules engine using BullMQ with exponential backoff retry; engineered CLI tunnel via WebSocket reverse-proxy to localhost without ngrok
- Shipped MCP server (`snag-mcp`) with 7 AI agent tools enabling Claude Code, Cursor, and Copilot to capture, inspect, and replay webhooks natively; published CLI (`snag-cli`) and SDK (`@snag/sdk`) to npm and PyPI with CI/CD via GitHub Actions

Tech: Next.js 15, Fastify, TypeScript, Python, WebSockets, Firebase Firestore, BullMQ, Upstash Redis, Ink, Zod, Tailwind CSS, shadcn/ui, Fly.io, Vercel

EDUCATION

B.Sc. Computer Science — University of Central Punjab, Lahore

Expected July 2026

Relevant Coursework: Machine Learning, Deep Learning, Data Structures, Algorithms, Database Systems

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, FastEmbed, SentenceTransformers, ONNX Runtime, LangChain, LlamaIndex, OpenAI/Anthropic APIs, ChromaDB, FAISS, Quantization (GGUF, GPTQ), LoRA/QLoRA, MLflow

Full-Stack: React, Next.js, TypeScript, Node.js, Express, .NET (C#), FastAPI, Flask, Electron, REST, GraphQL, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, Firestore, DynamoDB, Redis, Vector DBs (Pinecone, ChromaDB)

Cloud/DevOps: AWS (EC2, S3, Lambda, ECS), Docker, Kubernetes, GitHub Actions, Prometheus, Grafana

Tools: Git, Okta, Auth0, JWT, RabbitMQ, Celery, Jest, Pytest, Ruff, MyPy, Postman